The Uncertainty Principle

PHYS 2203 — Introductory Physics III

October 30, 2026

Overview

Last lecture introduced the Fourier transform and found that the Fourier transform of a Gaussian wave packet is another Gaussian. After a short recap, we compare the widths of the two Gaussians and find that the narrower the packet, the wider its range of wave numbers. The single slit shows the same behavior and reveals that diffraction is a Fourier transform. We then state the general inequalities $\Delta x\,\Delta k \geq \frac{1}{2}$ and $\Delta t\,\Delta\omega \geq \frac{1}{2}$, which hold for any wave. Finally, we define precisely what uncertainty means and, using $p = \hbar k$ and $E = \hbar\omega$, turn these inequalities into the uncertainty principles of quantum mechanics.

1 The Fourier Transform: A Recap

Last lecture we saw that any well-behaved nonperiodic function can be written as a sum of plane waves over a continuous range of wave numbers,

$$\psi(x) = \int_{-\infty}^{+\infty} A(k)\, e^{ikx}\, \mathrm{d}k, \qquad A(k) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \psi(x)\, e^{-ikx}\, \mathrm{d}x, \tag{1}$$

where the Fourier transform $A(k)$ is the amplitude of the plane wave with wave number k. For the Gaussian wave packet we found

$$\psi(x) = C\, e^{-(x/2\varepsilon)^2}\, e^{ik_0 x} \qquad \Longrightarrow \qquad A(k) = \frac{C\varepsilon}{\sqrt{\pi}}\, e^{-\varepsilon^2(k-k_0)^2}. \tag{2}$$

The Fourier transform of a Gaussian is another Gaussian, centered at k_0. With beats we already saw an inverse relationship between widths: two waves whose wave numbers differ by Δk form groups of length $\Delta x = 2\pi/\Delta k$. Today we derive this relationship for wave packets by comparing the widths of $\psi(x)$ and $A(k)$.

2 The Widths of a Wave Packet

How wide is the Gaussian packet, and how wide is the range of wave numbers needed to build it? A Gaussian with mean μ and standard deviation σ has the general form

$$G(x) \propto \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{3}$$

The mean μ is the center of the bell curve, and the standard deviation σ measures its width: about 68% of the area under the curve lies within σ of the mean.

The widths of a wave packet are measured from the probability densities, $|\psi(x)|^2$ and $|A(k)|^2$. Squaring the Gaussian packet and its Fourier transform,

$$|\psi(x)|^2 \propto \exp\left[-\frac{x^2}{2\varepsilon^2}\right], \qquad |A(k)|^2 \propto e^{-2\varepsilon^2(k-k_0)^2} = \exp\left[-\frac{(k-k_0)^2}{2\,(1/2\varepsilon)^2}\right]. \tag{4}$$

Comparing with the general form, $|\psi(x)|^2$ has mean 0 and standard deviation ε, and $|A(k)|^2$ has mean k_0 and standard deviation $1/2\varepsilon$. Calling these widths Δx and Δk,

$$\Delta x = \varepsilon, \qquad \Delta k = \frac{1}{2\varepsilon}, \qquad \boxed{\Delta x\,\Delta k = \frac{1}{2}} \tag{5}$$

The width of $\psi(x)$ is inversely proportional to the width of $A(k)$. A narrow packet (small ε) needs a wide range of wave numbers, and a nearly pure plane wave (large ε) needs only a narrow range (Figure 1).

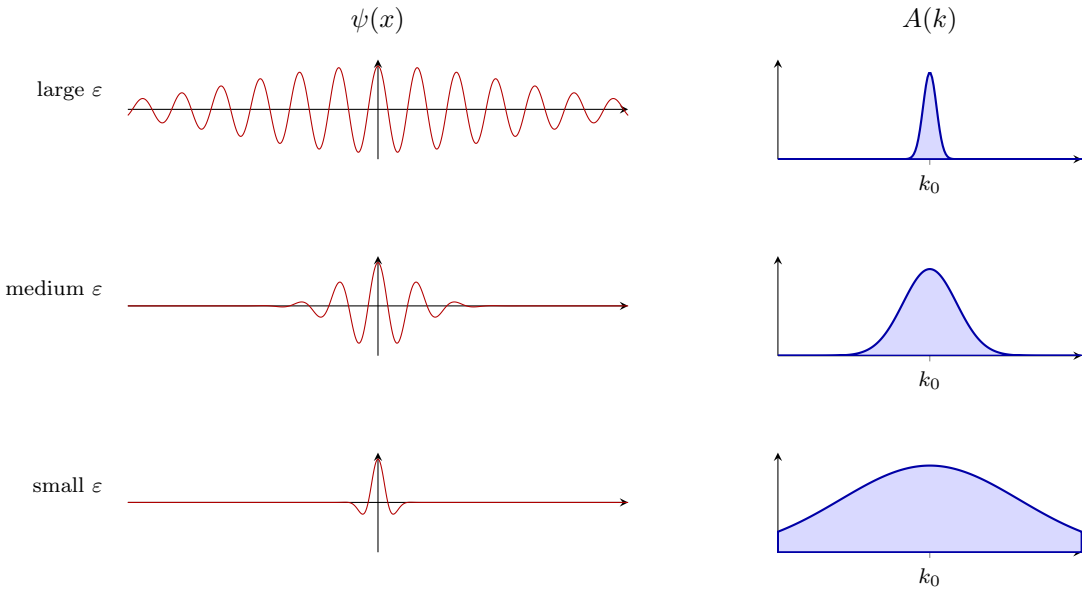


Figure 1: Gaussian packets of decreasing width (left) and their Fourier transforms (right). All have the same central wave number k_0. As $\psi(x)$ gets narrower, $A(k)$ gets wider.

3 Example: The Single Slit

Electrons of momentum p travel along z toward a slit of width w, which is narrow along x and long along y (Figure 2). Before the slit, all of the momentum is along z.

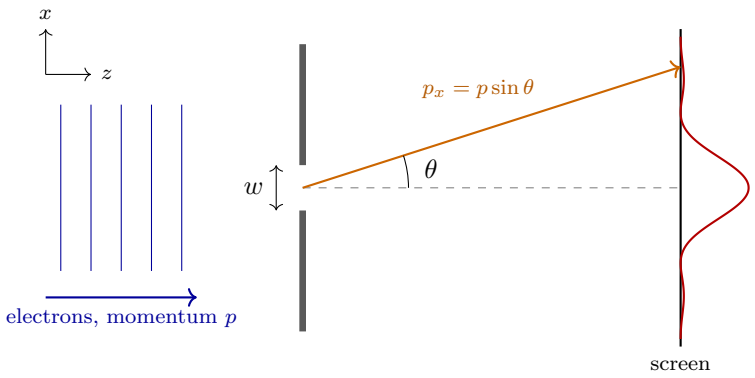


Figure 2: The single slit seen from above, looking along y. Electrons of momentum p arrive along z. Past the slit, an electron with sideways momentum p_x travels at an angle θ, with $\sin\theta = p_x/p$, and lands on a distant screen at a point set by θ. The curve on the screen is the intensity pattern.

Just past the slit, the wave function along x is roughly constant across the slit and zero outside (Figure 3, left):

$$\psi(x) = \begin{cases} C, & |x| < w/2 \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Its Fourier transform is

$$A(k) = \frac{1}{2\pi}\int_{-w/2}^{+w/2} C\left[\cos(kx) - i\sin(kx)\right] \mathrm{d}x = \frac{C}{2\pi}\,\frac{\sin(kw/2) - \sin(-kw/2)}{k} = \frac{C}{\pi}\,\frac{\sin(kw/2)}{k}. \tag{7}$$

The sine term integrates to zero because it is odd.

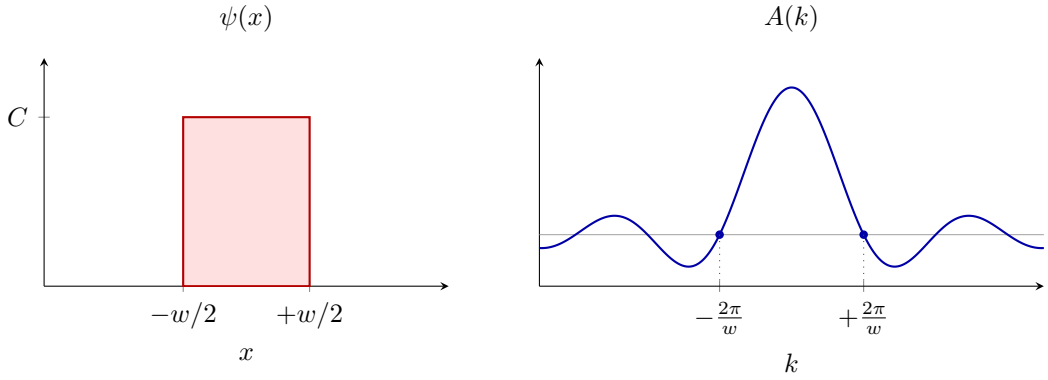


Figure 3: The single slit. Left: the wave function across a slit of width w. Right: its Fourier transform, whose first zeros (dots) are at $k = \pm 2\pi/w$. The gray line marks $A = 0$.

Here $\hbar k$ is the sideways momentum p_x, and an electron with this momentum leaves the slit at an angle θ with $\sin\theta = p_x/p = \hbar k/p$ (Figure 2). On a distant screen each angle receives one value of k, so the intensity pattern has the shape of $|A(k)|^2$, and the zeros of $A(k)$ are the diffraction minima. The first zero, at $k = 2\pi/w$, gives $\sin\theta = \lambda/w$, the single-slit minimum $w\sin\theta = \lambda$ of Lecture 9. In fact, adding the waves from all points across the slit, as we did in Lecture 9, is this same integral: the far-field diffraction pattern is the Fourier transform of the aperture. A narrower slit gives a wider $A(k)$, and so a wider pattern.

4 The Inverse Relationship

The Gaussian is a special case: it is the shape with the *smallest* product of widths. For any other function the product is larger:

$$\boxed{\Delta x\,\Delta k \geq \frac{1}{2}} \tag{8}$$

This is pure mathematics, true for any wave. Replacing x by t and k by ω gives the same relation in time:

$$\boxed{\Delta t\,\Delta\omega \geq \frac{1}{2}} \tag{9}$$

A signal that lasts only a short time must contain a wide range of frequencies.

Example: a short laser pulse. A laser pulse lasts $\Delta t = 10$ fs. Its frequency spread is at least

$$\Delta\omega \geq \frac{1}{2\Delta t} = \frac{1}{2(10\times 10^{-15}\text{ s})} = 5\times 10^{13}\text{ rad/s}, \qquad \Delta f = \frac{\Delta\omega}{2\pi} \geq 8\times 10^{12}\text{ Hz}. \tag{10}$$

For visible light ($f \approx 5\times 10^{14}$ Hz) this is a spread of a few percent: such a short pulse cannot be a single pure color.

5 What Uncertainty Means

Because of its wave nature, a particle's measured properties are inherently uncertain. Suppose an experiment is repeated many times *identically*, and a quantity Q (a position x, a momentum p_x, ...) is measured each time. The value Q_1 is obtained n_1 times, Q_2 is obtained n_2 times, and so on.

The mean.

$$\bar{Q} = \frac{\sum_i Q_i\, n_i}{\sum_i n_i} \tag{11}$$

The uncertainty. The uncertainty ΔQ is defined as the **standard deviation**: the square root of the mean of the squared deviations from the mean,

$$\boxed{\Delta Q = \sqrt{\frac{\sum_i (Q_i - \bar{Q})^2\, n_i}{\sum_i n_i}}} \tag{12}$$

$\Delta Q = 0$ only if every measurement gives the same value. The more spread out the results, the larger ΔQ. (Here Δ means uncertainty, not "change in".) When we speak of the uncertainty in an electron's momentum, we mean this specific number, not something vague.

6 The Uncertainty Principles

For a matter wave, $p = \hbar k$ and $E = \hbar\omega$. Multiplying the two inequalities of Section 4 by $\hbar$ turns statements about waves into statements about particles:

The uncertainty principles. Because of a particle's wave nature, its position and momentum along an axis cannot both be known precisely, and neither can its energy and the time over which it is measured:

$$\Delta p_x \, \Delta x \geq \frac{\hbar}{2}, \qquad \Delta E \, \Delta t \geq \frac{\hbar}{2}.$$

The first is the **Heisenberg uncertainty principle** (Heisenberg, 1932 Nobel Prize). These are not limitations of our instruments. They follow from the wave nature of matter.

* **Position and momentum.** A plane wave has a definite momentum ($\Delta p_x = 0$) but could be anywhere ($\Delta x = \infty$). Localizing the particle requires a spread of momenta, as in the single slit of Section 3, where confining the electron to $\Delta x \sim w$ spreads its sideways momentum. The relation holds separately along each axis, so there is no limit on knowing x and p_y together.

* **Energy and time.** A state that lasts only a time Δt has an uncertain energy. An excited state of an atom lives for a short time before emitting a photon, so its energy, and the photon's, is spread out, giving each spectral line a small natural width.

* **Size of the effect.** Because $\hbar$ is so small, the uncertainty principle is negligible for everyday objects but crucial for electrons in atoms.

* **Typical values.** If the average value of a quantity is zero, its typical values are about the size of its uncertainty. For an electron bound in an atom the average momentum is zero, so its typical momentum is about Δp. We will use this to estimate the size of the hydrogen atom in Lecture 36.